MITUL AADITHYA A.V.

Data Analytics | Data Engineering | Applied Machine Learning

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PROFESSIONAL SUMMARY

Computer Science undergraduate specializing in data analytics and applied machine learning, with hands-on experience in Python, SQL, and EDA through an NUS-mentored deep learning research project (LiverGuard) and academic AWS exposure (SageMaker, Glue, QuickSight). Builds structured, analytics-ready datasets and optimized SQL queries; adept at translating raw data into actionable insight under deadline pressure.

TECHNICAL SKILLS

Languages: Python, SQL, R

Data Engineering: ETL fundamentals, Hadoop, Spark, Airflow, Cron scheduling

Analytics & BI: Exploratory Data Analysis, Power BI, Business Analytics & Statistics

Cloud & Warehousing: AWS (Glue, SageMaker, QuickSight, Redshift) – academic exposure

Databases & Tools: MySQL, Git, VS Code, cURL testing

Operating Systems: Linux, Windows

PROJECTS

LiverGuard — AI/ML-Based Health Diagnosis System*NUS Global Academic Internship*

- Conducted EDA on a clinical dataset of 16,389 patient records, uncovering key correlations (e.g., 0.89 between total and direct bilirubin) and enzyme-marker patterns that informed feature selection
- Evaluated and compared ML algorithms (Random Forest: 99.94%, Decision Tree: 99.88%, KNN: 94.36%) for liver disease classification, contributing to an ensemble model that reached 99.91% accuracy with perfect precision/recall/F1 scores
- Analyzed and optimized a ResNet-50 deep learning pipeline for CT-scan liver segmentation and tumor detection, applying preprocessing (normalization, contrast enhancement) to improve robustness across scans

AttentionX — Personalized Attention Intelligence for E-Learning

- Built a real-time multimodal computer vision pipeline (OpenCV, MediaPipe Face Mesh, YOLOv8) that detected learner attention fluctuations with 92.8% average accuracy, outperforming gaze-only and emotion-only baseline models
- Fused visual (gaze, posture) and behavioral (tab focus, scroll, mouse) signals through a PyTorch adaptive neural network to compute real-time Attention, Fatigue, and Cognitive Load indices at sub-200ms latency
- Validated system output against manual ground-truth labeling, achieving 91% attention-state classification accuracy and 94% precision in distraction detection

GNS3 & RIP with IPv6 — Computer Networks Simulation

- Designed a multi-network topology using routing protocols, building system-level troubleshooting skills relevant to data infrastructure

Tennis Analysis System — Computer Vision for Sports Analytics

- Built a computer vision pipeline (YOLOv8, PyTorch) for real-time player and ball detection, paired with a custom CNN to extract 14 court keypoints per frame for full match analysis
- Implemented postprocessing techniques — classical CV-based keypoint refinement and homography correction — improving precision from 0.936 to 0.963 and cutting median keypoint error from 2.83px to 1.83px

- Trained and validated models on a curated dataset of 8,841 annotated tournament-video frames (75/25 train-validation split)

INTERNSHIP

Global Academic Internship Programme (GAIP), National University of Singapore *Dec 2024 – Jan 2025*

- Completed an intensive research project on Big Data Analytics with Deep Learning under NUS School of Computing faculty mentorship, contributing EDA and DL algorithm analysis to a 6-person team building the LiverGuard diagnosis system
- Gained direct exposure to AWS services (SageMaker, Glue, QuickSight) for data processing and visualization, collaborating with an international peer cohort

CERTIFICATIONS

- **Big Data Analytics using Deep Learning** — NUS School of Computing, Jan 2025
- **Deep Learning with Amazon SageMaker** — NUS & AWS Training and Certification, Jan 2025
- Advanced Relational Databases & SQL — Coursera
- NPTEL Certifications — scores 100%, 88%
- Modern Web Development with MERN Stack — FACE Prep, May–Jun 2025

EDUCATION

Vellore Institute of Technology (VIT), Chennai — B.Tech, Computer Science & Engineering *CGPA: 8.05*

DAV Public School, Velachery, Chennai *Class XII: 91% (2022) | Class X: 90.8% (2020)*

EXTRACURRICULAR ACTIVITIES

CYSCOM — Cybersecurity Club, VIT Chennai *Aug 2023 – Apr 2024*

- Participated in and hosted cybersecurity workshops and hacking challenges (TryHackMe, Hack The Box), building systems security fundamentals